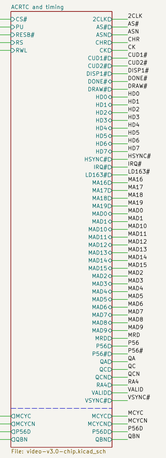
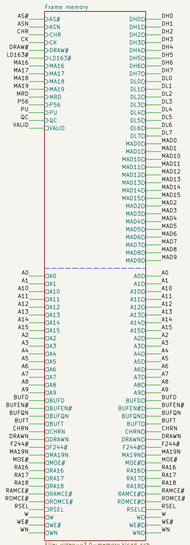
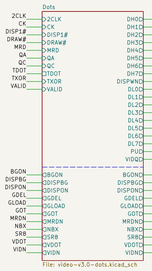
The video card V3.0, one sheet per functional block:
  sheet 2 Bus and ports BUS INTERFACE: THE SIX PORTS' STROBES, THE CHIP'S CS, RS AND R/W, ITS DATA BUFFER
  sheet 3 ACRTC and timing THE HD63484 ACRTC, THE DOT CLOCK, THE DOT COUNTER, PS6 AND VALID
  sheet 4 Frame memory ADDRESS LATCHES, THE TWO RAMS, THE FONT.EPROM AND ITS COPY, THE MAD BUFFERS, THEIR CONTROL
  sheet 5 Dots   THE GRAPHICS SHIFT REGISTERS AND THEIR LOAD, THE DELAY LINE, THE TWO LAYERS MIXED, THE DOT RETIMED
  sheet 6 Control and video THE CONTROL REGISTER FROM THE ATTRIBUTE BYTE, THE THREE GUNS, THE SYNCS, THE VGA SOCKET
  sheet 7 Text A   TEXT METHOD A: EIGHT SOCKETS, EMPTY ON THE FIRST BUILD
  sheet 8 PS2   PS/2: ATMEGA328P. LATCHES ON PE / PF. THE CARD STATUS, FTDI. –INT, RESET
  sheet 9 Power   POWER UNITS, DECOUPLING; POWER LED, THE F PARTS' PULL-UP

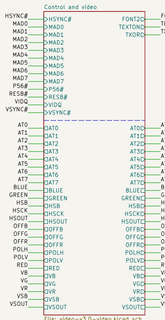
Every net of a sheet is a pin of its sheet symbol here, and the local label at each pin names the net
(/NAME, the names on the board). Above the dashed line: the nets the sheet shares with other sheets (left:
it reads them; right: it drives them). Below it: the nets drawn on that sheet only, a pin on each side
(KiCad's ERC counts a root label as connected only when another label of its name is on the root sheet).
VCC and GND are power symbols on the sheets.

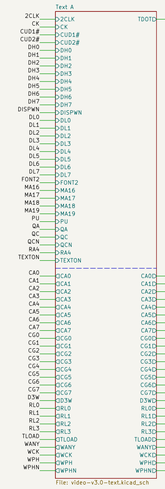


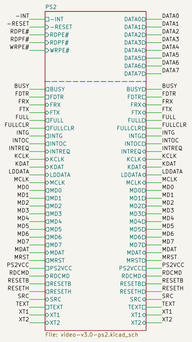














Design note and theory: docs/cards/video–v3.md
Circuit source: v30_netlist.py: these sheets are generated by gen_v30.py
HD63484 ACRTC on ports PA–PD, 640 x 480 VGA, sockets for text method A, PS/2 keyboard and mou
YACC1

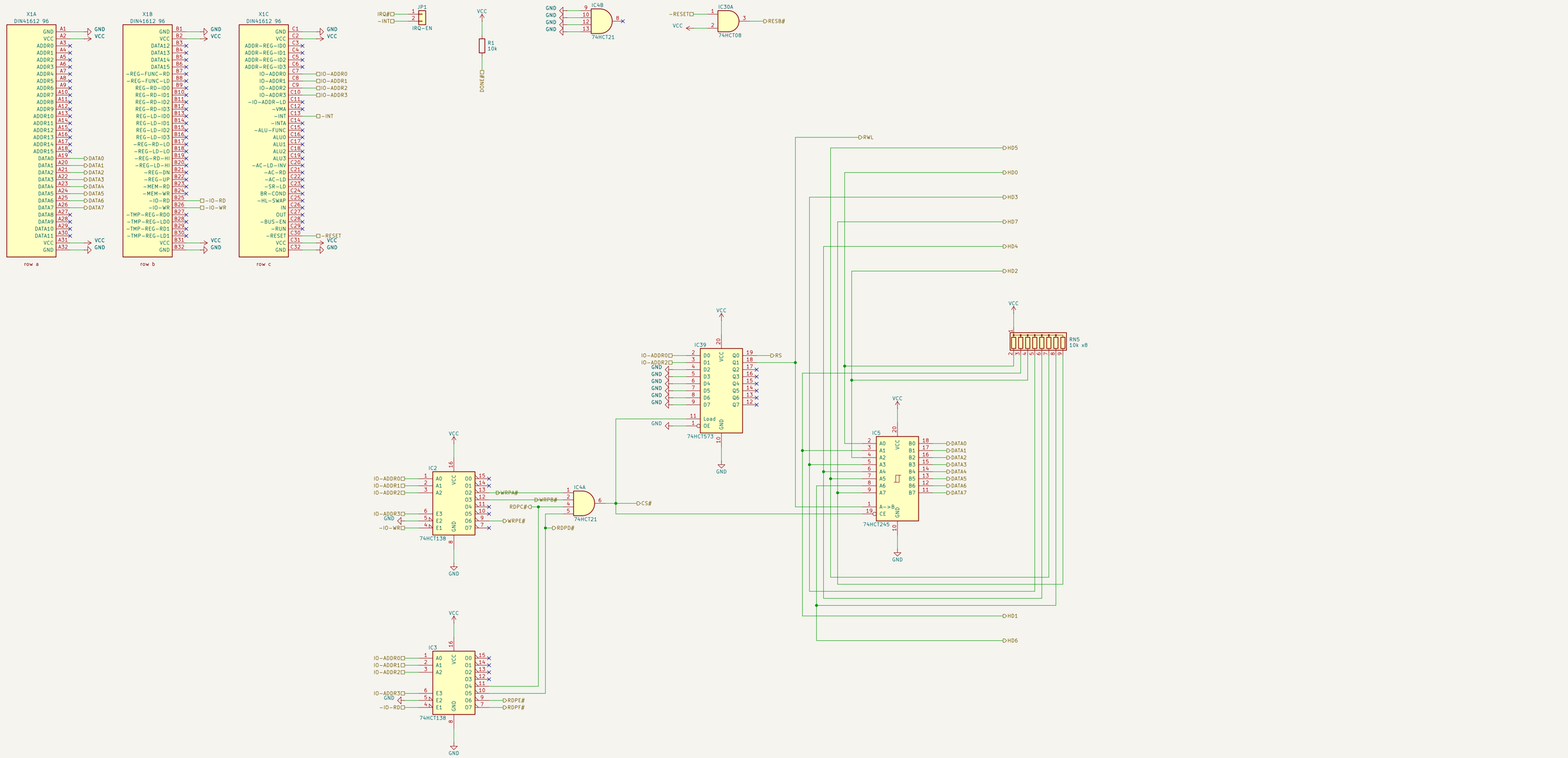
Sheet: /
File: video–v3.0.kicad_sch

Title: YACC1 video card V3.0

Size: A1	Date: 2026–11–11	**Rev: V3.0**
KiCad E.D.A. 10.0.6		Id: 1/9

BUS INTERFACE: THE SIX PORTS' STROBES, THE CHIP'S CS, RS AND R/W, ITS DATA BUFFER

X1: the YACC1 V3.2 bus (DIN 41612). IC2 (writes) and IC3 (reads) decode IO-ADDR0-3: ports PA-PF = 1010-1111.
CS (IC4) = any of OUT PA, OUT PB, IN PC, IN PD. RS = IO-ADDR0 and R/W = IO-ADDR2 through IC39, a latch that CS closes: an IN record drops the port number with its strobe, and the chip wants them until CS has risen.
IC5 between the chip's D0-7 and the bus: enabled by CS, direction R/W. IC30/A: -RESET to the chip's RES and DACK.



Design note and theory: docs/cards/video-v3.md
Circuit source: v30_netlist.py; this sheet is generated by gen_v30.py
Sheet 2 of the video card V3.0 schematic (root: video-v3.0.kicad_sch)

YACC1

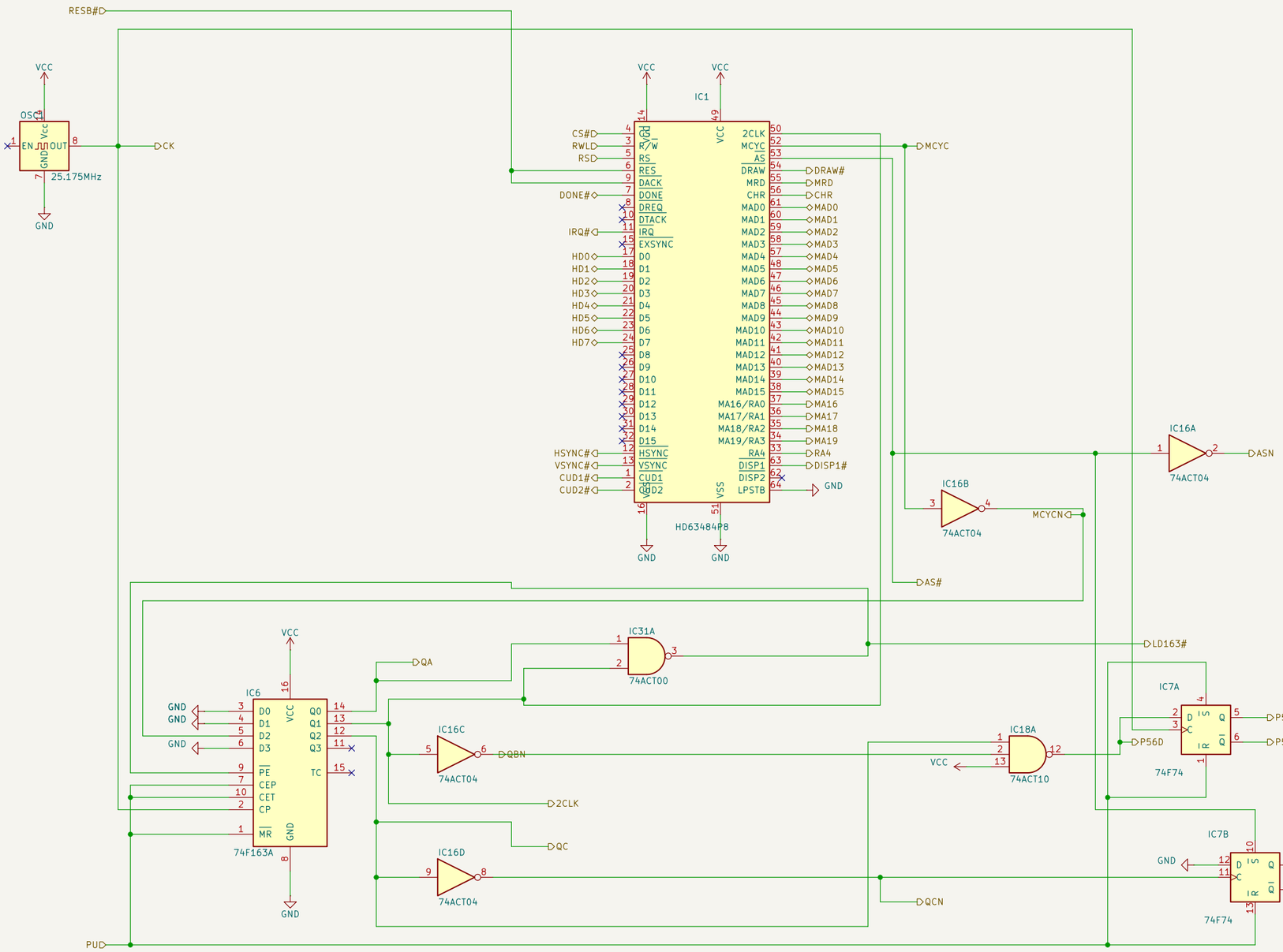
Sheet: /Bus and ports/
File: video-v3.0-bus.kicad_sch

Title: YACC1 video card V3.0: Bus and ports

Size: A2	Date: 2026-11-11	**Rev: V3.0**
KiCad E.D.A. 10.0.6		Id: 2/9

THE HD63484, THE DOT CLOCK, THE DOT COUNTER, P56 AND VALID

IC6 counts the 25.175 MHz dot clock: QB is the chip's 2CLK, QC follows its MCYC (the counter loads NOT MCYC in dots 3 and 7, so QA–QC are the dot number in a memory cycle). IC7/A is P56, a flip-flop high in dots 5 and 6 of every cycle: the one edge the card times things with. IC7/B is VALID: AS came in this cycle (a dummy cycle has none). The chip's host side is on sheet 2, its memory side on sheets 4 to 7.



Design note and theory: docs/cards/video-v3.md
Circuit source: v30_netlist.py; this sheet is generated by gen_v30.py
Sheet 3 of the video card V3.0 schematic (root: video-v3.0.kicad_sch)
YACC1

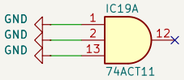
Sheet: /ACRTC and timing/ File: video-v3.0-chip.kicad_sch		
Title: YACC1 video card V3.0: ACRTC and timing		
Size: A3	Date: 2026-11-11	**Rev: V3.0**
KiCad E.D.A. 10.0.6		Id: 3/9

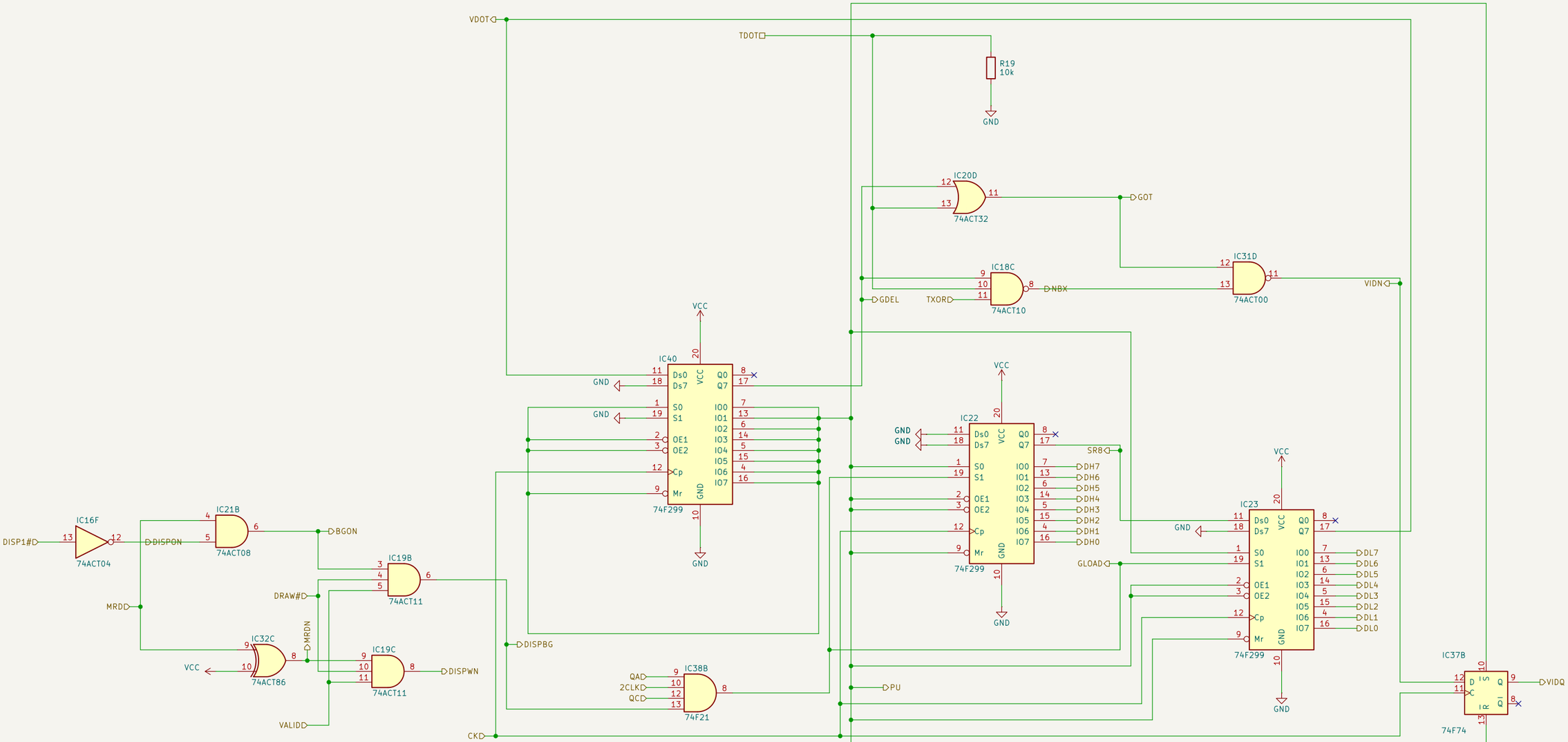
ADDRESS LATCHES, THE TWO RAMS, THE FONT EPROM AND ITS COPY, THE MAD BUFFERS, THEIR CONTROL

IC8 / IC9 hold MAD0–15 while AS is high. The RAM where MA19 is low or CHR is high, the font (IC14, its byte on both halves through IC15) where MA19 is high; CHR (a character screen) forces A16–18 high (IC36). MOE (IC18/8) = NAND(VALID, AS, NOT W): the memories drive the data lines from the end of AS to the end of the cycle, never in a drawing write (W, IC17/A). ~WE = NAND(P56, W). IC12 / IC13 between the RAM and MAD in a drawing cycle: IC37/A, on in dots 5–7 of a write and 6–7 of a read. JP4: pin 30 of the RAMs, CE2 or A17.

THE GRAPHICS SHIFT REGISTERS AND THEIR LOAD, THE DELAY LINE, THE TWO LAYERS MIXED, THE DOT RETIMED

IC22 / IC23 load the RAM's word in dot 7 of a Background cycle with the display on (GLOAD, IC38/B) and shift it out bit 0 first; a word fetched with the display off is not loaded, which is the blanking. IC40 delays the dot 8 dots, into step with the text. VIDN = NOT ((G OR T) AND NOT (G AND T AND TXOR)); IC37/B puts it on the dot clock. DISPWN (IC19/C) marks a Window cycle for method A. R19 holds the text dot low without its parts.



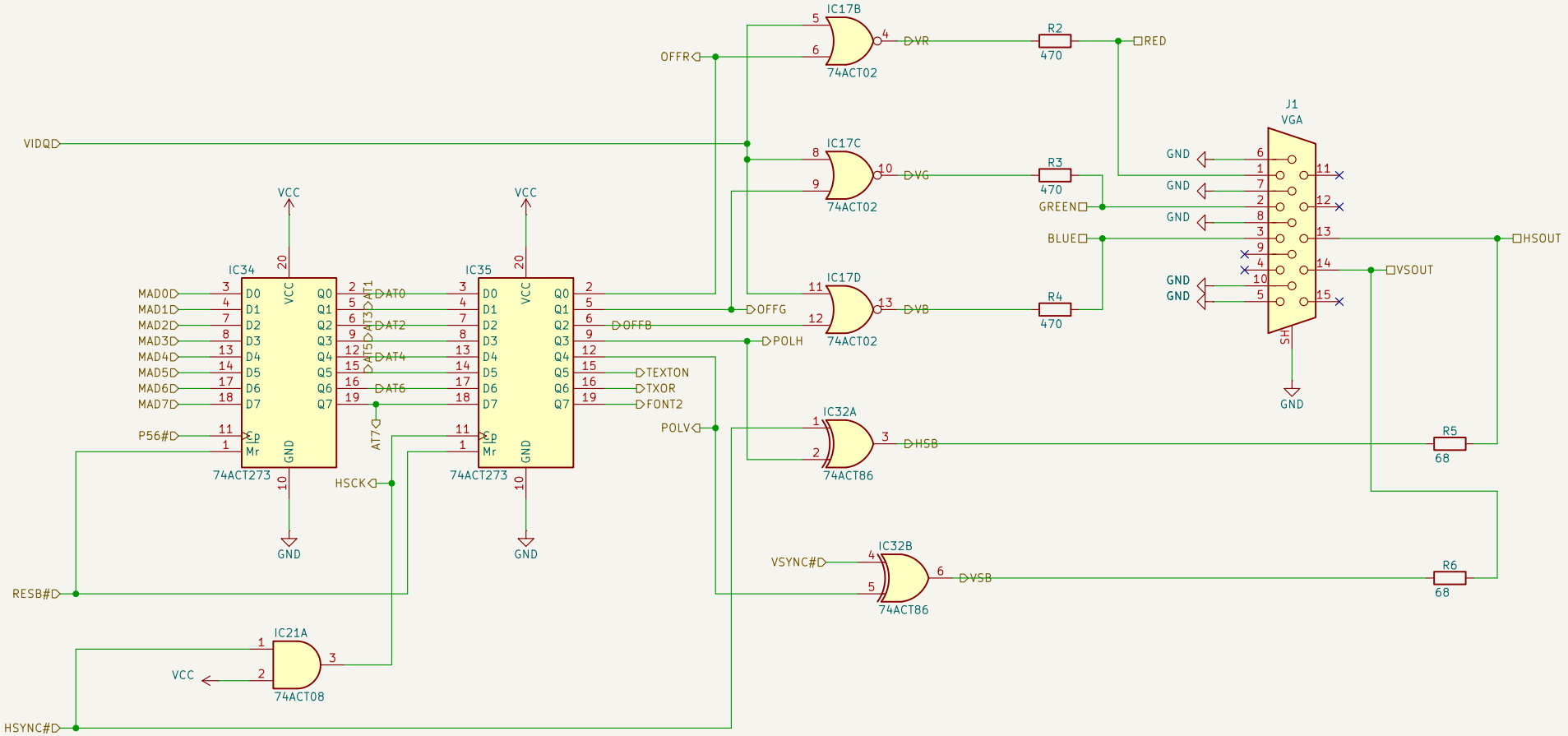


Design note and theory: docs/cards/video-v3.md
Circuit source: v30_netlist.py; this sheet is generated by gen_v30.py
Sheet 5 of the video card V3.0 schematic (root: video-v3.0.kicad_sch)
YACC1

Sheet: /Dots/ File: video-v3.0-dots.kicad_sch		
Title: YACC1 video card V3.0: Dots		
Size: A3	Date: 2026-11-11	**Rev: V3.0**
KiCad E.D.A. 10.0.6		Id: 5/9

THE CONTROL REGISTER FROM THE ATTRIBUTE BYTE, THE THREE GUNS, THE SYNCS, THE VGA SOCKET

IC34 takes MAD0–7 as dot 7 begins in every cycle (P56's end); IC35 copies it when HSYNC rises, the one moment it holds the chip's attribute byte (ATR). Bits 0–2: red, green, blue off; 3, 4: HSYNC, VSYNC positive; 5: text on; 6: XOR; 7: the second font. Each gun = the dot AND NOT its off bit, through 470 ohm into the monitor's 75.



Design note and theory: docs/cards/video-v3.md
Circuit source: v30_netlist.py; this sheet is generated by gen_v30.py
Sheet 6 of the video card V3.0 schematic (root: video-v3.0.kicad_sch)
YACC1

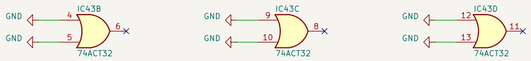
Sheet: /Control and video/
File: video-v3.0-video.kicad_sch

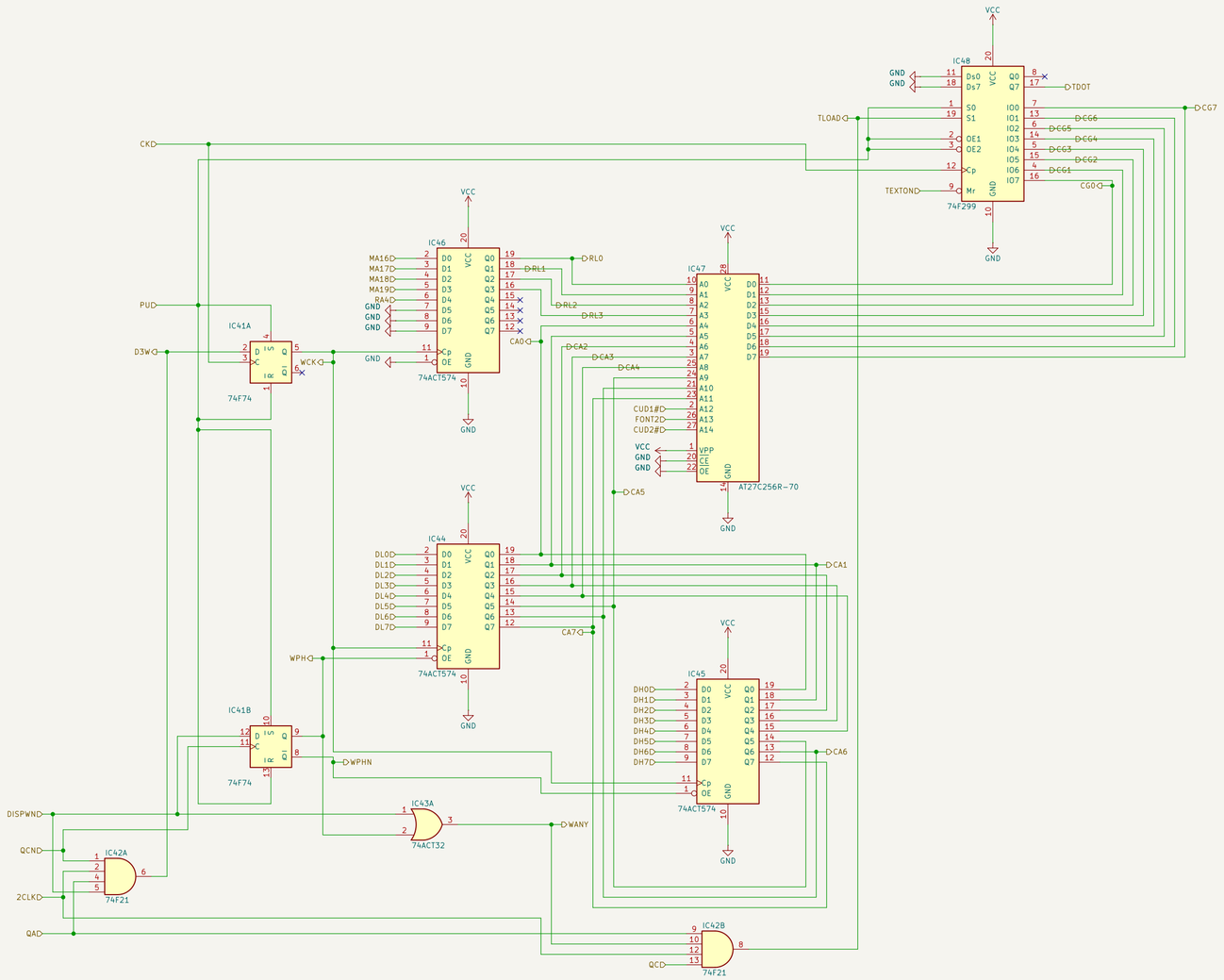
Title: YACC1 video card V3.0: Control and video

Size: A4	Date: 2026-11-11	**Rev: V3.0**
KiCad E.D.A. 10.0.6		Id: 6/9

TEXT METHOD A: EIGHT SOCKETS, EMPTY ON THE FIRST BUILD

In superimposed access the Window's cycle reads a character word: IC44 / IC45 take its two codes and IC46 the raster line as dot 4 begins (WCK, IC41/A). The left code addresses the character generator IC47 for the rest of the cycle, the right one through the next (WPH, IC41/B); IC48 loads the 8 dots at each cycle's end and shifts them out as TDOT. CUD1, CUD2 and FONT2 are address lines of IC47: the cursor and a second font.





Design note and theory: docs/cards/video-v3.md
Circuit source: v30_netlist.py; this sheet is generated by gen_v30.py
Sheet 7 of the video card V3.0 schematic (root: video-v3.0.kicad_sch)
YACC1

Sheet: /Text A/
File: video-v3.0-text.kicad_sch

Title: YACC1 video card V3.0: Text A

Size: A2	Date: 2026-11-11	**Rev: V3.0**
KiCad E.D.A. 10.0.6		Id: 7/9

PS/2: ATMEGA328P, LATCHES ON PE / PF, THE CARD STATUS, FTDI, −INT, RESET

IC24 runs embedded/video-ps2. OUT PE loads a command into IC26 (BUSY set); IN PF reads a byte from IC25 (FULL cleared). IN PE the status IC27: FULL SRC. BUSY. TEXT (JP3: method A's ports are fitted) and the ID 0110. −INT = INTREQ AND FULL (IC30/B) through IC29/D and JP2. The bus −RESET reaches MRST through IC30/C, IC29/E and IC29/F (open collector, RB): it only ever pulls MRST low.

IC29A 7416 · IC29B 7416 · IC29C 7416

RDPF#D · IC30D 74HCT08 · −RESETD · FULLCLR

IC26 74HCT374 — DATA0..DATA7 · MD0..MD7 · RDCMD

IC24 ATmega328P-P — AREF · PB0..PB5 · XTAL1/PB6 · XTAL2/PB7 · PC0..PC6 · RESET/PC6 · PD0..PD7 · MD6 · FTX · XCLK · MCLK · KDAT · MDAT · INTREQ · BUSY · MRST · CFDTR · FRX

Y1 16MHz · C53 22p · C54 22p · R11 10k

J1 FTDI

IC28A 74HCT74 — VCC · OLDDATA · FULL

IC30B 74HCT08 — SRC · INTG

IC29D 7416 — INTGC · −INT · JP2 KBD-INT

IC25 74HCT374 — DATA0..DATA7

IC27 74HCT244 — DATA0..DATA7 · TEXT

R12 4.7k · R13 4.7k · R14 4.7k · R15 4.7k

J2 KEYBOARD · J3 MOUSE · PS2VCC

IC28B 74HCT74

C55 100n · R16 10k · JP3 TEXT

R9 10k · R10 10k · R7 10k

WRPE#D · RDPE#D

IC30C 74HCT08 — −RESETD · RESETB

IC29E 7416 · IC29F 7416 · RESETH

R8 4.7k

F1 0.5A PTC

Design note and theory: docs/cards/video-v3.md
Circuit source: v30_netlist.py; this sheet is generated by gen_v30.py
Sheet 8 of the video card V3.0 schematic (root: video-v3.0.kicad_sch)

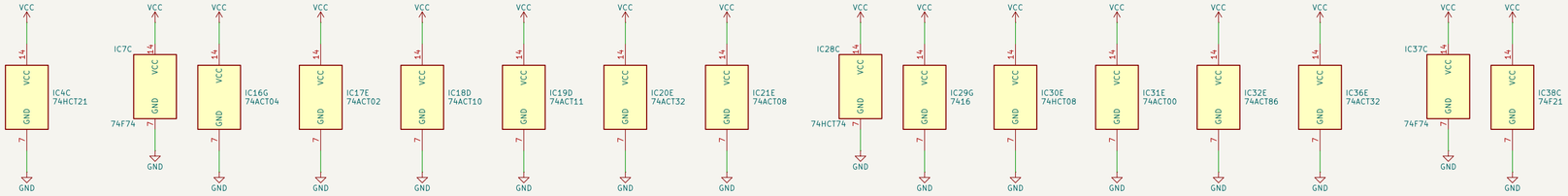
YACC1
Sheet: /PS2/
File: video-v3.0-ps2.kicad_sch

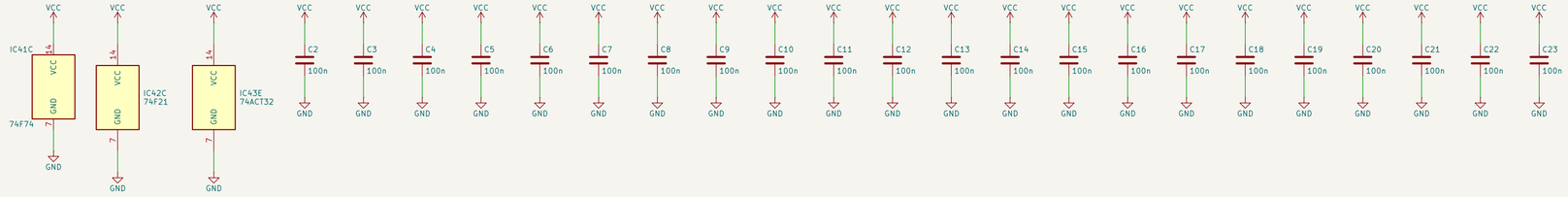
Title: YACC1 video card V3.0: PS2

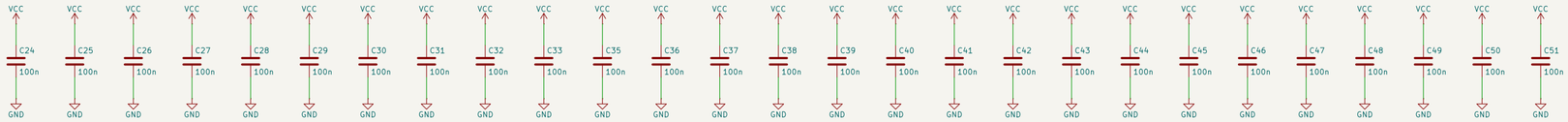
Size: A1	Date: 2026-11-11	Rev: V3.0
KiCad E.D.A. 10.0.6		Id: 8/9

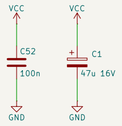
POWER UNITS, DECOUPLING, POWER LED, THE F PARTS' PULL-UP

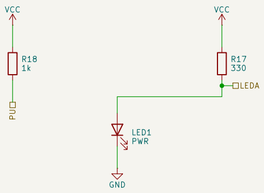
One 100 nF per IC beside its VCC pin (C50 at the chip's second VCC pin, C52 at the ATmega's AVCC). C1 47 uF bulk. The bus supplies VCC and GND (X1, six pins each): the PWR_FLAGs. PU (R18, 1k) holds the 74F parts' unused inputs high. Spare gates: IC4/B, IC19/A, three of IC29, three of IC43.













PWR_FLAGs: the bus supplies VCC and GND (X1, six pins each).

Design note and theory: docs/cards/video-v3.md
Circuit source: v30_netlist.py; this sheet is generated by gen_v30.py
Sheet 9 of the video card V3.0 schematic (root: video-v3.0.kicad_sch)

YACC1

Sheet: /Power/
File: video-v3.0-power.kicad_sch

Title: YACC1 video card V3.0: Power

Size: A2	Date: 2026-11-11	**Rev: V3.0**
KiCad E.D.A. 10.0.6		Id: 9/9